

Hassan Elhllos

Embedded Systems & PCB Design Engineer

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PROFESSIONAL SUMMARY

Embedded Systems & PCB Design Engineer with 3+ years of freelance experience building custom IoT hardware end-to-end - from schematic design and industrial PCB layout to firmware development (C/C++) and cloud integration. Proficient with ESP32, STM32, KiCad and Altium Designer, and hands-on with industrial automation (Siemens S7-1200, Schneider PLC, VFD, Modbus RS485). Awarded 1st place at Suez Canal University's Canal Startup Sprint for an industrial PLC product (ESP32).

TECHNICAL SKILLS

Embedded Systems: ESP32, STM32, ESP12E/F, Arduino, Raspberry Pi, FreeRTOS concepts, GPIO, Timers, Interrupts
PCB & Hardware Design: KiCad, Altium Designer, Schematic Design, PCB Routing, EMC-hardened layout, Proteus, Digital Fabrication
Industrial Automation: Siemens S7-1200 PLC, Schneider PLC, Ladder Logic, Modbus RTU (RS485), VFD configuration, Classic Control (star-delta, overload protection)
Software & Data: C/C++, Python, SQL, Git/GitHub, JSON, Embedded firmware, Ladder Diagram, Dashboard development
Connectivity: MQTT (HiveMQ), UART, SPI, I2C, RS485, HTTP, ESP RainMaker, Blynk, Wi-Fi, BLE
Computer Vision: OpenCV, Object Detection, Image Processing on Raspberry Pi / embedded platforms
Hardware Lab: Oscilloscope, Multimeter, Soldering (through-hole & SMD), USB-to-UART, EMI troubleshooting

EXPERIENCE

Independent IoT & Hardware Developer 2023 - Present
Freelance

- Design custom IoT prototypes and PCB layouts for academic and client projects, integrated with cloud platforms (ESP RainMaker, Blynk) for remote control and rapid deployment.

Industrial Automation Trainee Sep 2025 - Nov 2025

HA Consulting Group

- Configured VFDs and designed classic control circuits (star-delta starter, overload protection).
- Programmed PLC logic on Siemens S7-1200 and Schneider TIA / CoDeSys platforms.

IoT Intern May 2025 - Jun 2025

Creativa Innovation Hub (ITIDA) - Grade: Excellent

- Configured ESP32 edge platforms with reliable MQTT connectivity (HiveMQ broker).
- Built cloud dashboards for real-time IoT monitoring; architected a Smart Agriculture system with edge-to-cloud MQTT communication.

Computer Vision Volunteer 2024 - 2025

SCU Racing Team (ROV) - Suez Canal University

- Developed underwater object detection and navigation modules in Python & OpenCV on Raspberry Pi.

Data Analysis Intern Jul 2024 - Aug 2024

Information Technology Institute (ITI)

- Applied SQL and data-cleaning techniques to extract insights from real-world datasets, then visualized them in dashboards.

SELECTED PROJECTS

Custom PLC - Industrial Controller: industrial-grade PCB hardened for EMC, 8 digital inputs / 8 outputs, Modbus to HMI/SCADA, custom Ladder-programming IDE, 3D-printed enclosure. *1st Place, Canal Startup Sprint.*

Custom Smart Home System: ESP32-based controller with 8-channel relay on a custom KiCad PCB; remote access via ESP RainMaker / Blynk; schedules and scenes.

Cold Storage Monitoring & Control: Dual ESP32 boards monitoring industrial temperature sensors and fault-lamp signals from large fruit cold-storage panels, with remote control and instant fault alerts.

ESP32 DevKit Board: Custom KiCad board with built-in auto-programming (one-wire) circuit replacing the BOOT button, matching the original 2x20 pinout.

Star-Delta Starter Control Panel: Classic motor-control panel (contactors, timer, overload relay) for safe 3-phase induction-motor starting.

EDUCATION

B.Sc. in Electrical Engineering 2021 - 2026
Faculty of Engineering - Suez Canal University

Diploma, Advanced Technical School for Information Technology 2016 - 2021
Ismailia, Egypt

Advanced Industrial Electronics Diploma Ongoing
Digital Fabrication - High-Speed PCB Design & SMPS

HONORS & CERTIFICATION

1st Place - Canal Startup Sprint, Suez Canal University (graduation project: Industrial Controller ESP32 PLC).

Certification - Industrial Automation - HA Consulting Group (Sep - Nov 2025).

Certification - IoT Internship - Creativa Innovation Hub (ITIDA), Grade: Excellent.

Certification - Data Analysis - Information Technology Institute (ITI).

LANGUAGES

Arabic Native
English Professional